# Foveated Variable-Resolution Path Tracing in Cycles

## Design document — desktop display target

**VISGRAF / IMPA — Foveal Stereo Vision project**
Draft 1, September 2026. Companion to *Physiology of Foveated Human Vision* and *Eye movements and stereoscopic depth perception*.

---

## 0. Summary of design decisions

| Decision | Choice | Section |
|---|---|---|
| Display target | 27" 4K desktop, 600 mm, 67.3 px/deg on axis | §1 |
| Perceptual law | Geisler–Perry contrast-threshold cutoff, not MAR acuity | §4 |
| Quality knob | $CT_0$ (a threshold contrast), single parameter | §4.3 |
| Primary-ray density | $D(e) = \min(1, 2f_c(e)/N_d)^2 \cos^3\theta$, floored | §4.4 |
| Per-site path count | Derived by constrained allocation, not tuned | §5 |
| Lattice | Row-structured (scanline-stratified), not free blue noise | §6 |
| Reconstruction | In sampling space, denoise before upsampling | §7 |
| Cycles hook | Path-index→pixel map + per-pixel sample-count buffer | §9 |
| Declared budget | $B = 0.054$ of full-resolution primary rays | §2 |
| Headline claim to test | End-to-end 5–20×, depending on one parameter $\gamma$ | §5.4 |

The single most important quantitative finding in this document is in §5.4: **the periphery may need *more* paths per surviving sample, not fewer**, and the parameter that decides this swings the achievable speedup by a factor of four. Everything else is comparatively settled.

---

## 1. Scope and reference configuration

### 1.1 What this system is

An offline path tracer that spends its ray budget according to where a viewer is looking, producing images that are perceptually indistinguishable from a uniformly-rendered reference at that fixation. The renderer is Cycles. The output is a still image or a stereo pair, not a real-time stream.

This is a different regime from the real-time VR foveation literature and the difference is favourable. Real-time foveated path tracers operate at 1 spp and are dominated by reconstruction and denoising overhead; an offline renderer running for tens of seconds to minutes per frame amortises those fixed costs to near zero. We should therefore expect to get closer to the theoretical bound than Koskela et al.'s reported 2.5×.

## 1.2 What this system is not

Not a real-time renderer. Not gaze-contingent display (no tracker in the loop for the first two stages; gaze is a specified parameter). Not a peripheral-appearance model (no metamers, no texture synthesis in the periphery). Light transport is held fixed; only sampling varies.

## 1.3 Reference display

All numbers in this document assume:

| Quantity | Value |
|---|---|
| Panel | 3840 × 2160, 27" 16:9 |
| Physical size | 597.7 × 336.2 mm (6.424 px/mm) |
| Viewing distance $D$ | 600 mm |
| **On-axis angular pixel density $N_d$** | **67.3 px/deg** |
| Display Nyquist | 33.6 cycles/deg |
| Field of view | 53.0° × 31.3° |
| Max off-axis angle $\theta$ (corner) | 29.75° |
| Max eccentricity (gaze at opposite corner) | 48.8° |

$N_d$ is the quantity that connects the perceptual model to the renderer, and its absence is why most informal treatments of this problem cannot be checked. It is computed as

$$N_d = \frac{\text{px}}{\text{mm}} \cdot \frac{\pi D}{180}.$$

Everything below is stated relative to $N_d$ so that changing the display or viewing distance changes one number.

## 1.4 The desktop regime is qualitatively different from HMD

Three consequences follow from the table above and they shape the whole design.

**The display is roughly matched to foveal acuity.** At 33.6 cpd Nyquist against a foveal cutoff near 39 cpd, the fovea is display-limited, not perception-limited. There is nothing to gain in the fovea and no need for a supersampled baseline. Contrast this with an HMD at 20–35 px/deg, where the display is far coarser than the fovea and the honest baseline is a supersampled render.

**The eccentricity range is small.** A 53° horizontal FOV means eccentricity rarely exceeds 30°, versus 50–60° in a wide-FOV headset. Foveation gain grows with eccentricity range, so the desktop ceiling is lower than VR papers suggest.

**Viewing distance is the dominant unknown.** Moving from 600 mm to 450 mm changes $N_d$ by 25% and changes every derived quantity. The system must take $D$ as an explicit input and must not have a hard-coded pixel-space falloff.

---

# 2. Declared budget

The project's framing is *the same posterior for less computation*. That is only a claim if the computation is declared. It is declared here.

**Primary-ray budget $B$.** The fraction of full-resolution primary camera rays the renderer is permitted to trace, averaged over the frame:

$$B = \frac{1}{|\Omega|} \int_\Omega D(e(x))\, dx.$$

For the reference display, the recommended operating point (§4.4) gives

$$\boxed{B = 0.054} \qquad \text{i.e. a } 18.6\times \text{ reduction in primary rays.}$$

**Total path budget $T$.** The fraction of full-resolution total paths, which folds in per-site path counts. This is the number that matters for wall-clock time and it is *not* $B$; see §5.4.

**Baseline.** Equal-quality comparison against a uniform render at 3840 × 2160 with Cycles adaptive sampling at noise threshold $\tau_0$ and the same denoiser configuration. Equal-time comparison is reported secondarily. Any speedup claimed without naming which of the two is being used is meaningless.

**Reporting rule.** Every performance claim in the resulting report states $D$, $N_d$, the gaze point, $CT_0$, the floor, and $\gamma$. Without those six numbers the result is not reproducible.

---

# 3. Evaluation protocol

Fixed before the method, because it determines the parameters.

## 3.1 Objective metric

**FovVideoVDP** (Mantiuk et al. 2021) as the primary metric, configured with the reference display parameters above. It is the only widely-used metric that takes eccentricity as an input, and it handles both spatial and temporal degradation. HDR-VDP-3 as a secondary check for the still-image case. PSNR and SSIM are reported only because reviewers expect them; neither is eccentricity-aware and neither should drive any decision.

Report JOD (just-objectionable-difference) against the uniform reference. Target: **JOD degradation ≤ 1.0** at the declared budget.

## 3.2 Psychophysical validation

The objective metric calibrates the model; a small experiment validates it.

- **Task:** two-interval forced choice, foveated vs. uniform render of the same scene, gaze enforced by a fixation cross and verified by an eye tracker (trials with fixation breaks > 1° are discarded).
- **Design:** adaptive staircase on $CT_0$, converging to the threshold value at which discrimination reaches 75%.
- **N:** 12–15 observers, 6–8 scenes chosen to span the failure modes (glossy interior, foliage, caustics, large flat gradients, high-frequency texture).
- **Output:** the threshold $CT_0$ and its confidence interval. The system then ships at a conservative fraction of it.

The staircase is the deliverable that makes this research rather than engineering. Without it, every parameter in §4 and §5 is a guess.

## 3.3 Failure-mode battery

Fixed scenes that specifically probe known foveation failures, rendered and inspected at every milestone:

1. Large low-frequency luminance gradient (probes peripheral blotching from denoiser + sparse sampling)
2. Fine repeating texture crossing the fovea/periphery boundary (probes aliasing at the density transition)
3. Peripheral glossy highlight (probes the variance/perception interaction of §5)
4. Peripheral caustic (probes whether the variance estimator warms up at all in the periphery)
5. Rapid gaze relocation across a high-detail region (probes hysteresis, §8)

---

# 4. Perceptual model

## 4.1 Why not the MAR / $E_2$ acuity law

The minimum-angle-of-resolution law $s(e) = s_0(1 + e/E_2)$ predicts when a high-contrast target becomes unresolvable. Rendering error is not a high-contrast target. It is broadband, low-to-mid contrast, and its visibility is governed by the contrast sensitivity function, not by acuity.

There is also a practical problem: $E_2$ is not one number. It is roughly 2.3–3° for acuity, 0.4–1° for crowding, and different again for contrast detection and for motion. Choosing "the" $E_2$ silently commits to a task the renderer is not performing.

## 4.2 The model actually used

Geisler & Perry (1998), contrast threshold as a function of spatial frequency and eccentricity:

$$CT(f, e) = CT_0 \exp\!\left(\frac{\alpha f (e + e_2)}{e_2}\right).$$

Setting $CT = 1$ (the maximum achievable contrast) gives the cutoff frequency beyond which nothing is visible at eccentricity $e$:

$$f_c(e) = \frac{e_2 \ln(1/CT_0)}{\alpha\,(e + e_2)} = \frac{f_c(0)}{1 + e/e_2}.$$

Note the form. This is the *same hyperbolic falloff* as the MAR law, so nothing is lost geometrically. What is gained is threefold: $f_c$ comes out in **absolute cycles per degree**, so it can be compared against $N_d/2$; $CT_0$ appears as an explicit, interpretable quality knob; and $\alpha$ and $e_2$ are contrast-domain parameters with published values rather than a task-ambiguous $E_2$.

With the standard $\alpha = 0.106$, $e_2 = 2.3°$, $CT_0 = 1/64$:

$$f_c(0) = 39.2 \text{ cpd}.$$

Against a display Nyquist of 33.6 cpd, **the fovea is display-limited**, and the perceptual limit only begins to bind beyond $e \approx 0.4°$. On this display the "fovea" in the rendering sense is about half a degree wide.

## 4.3 The quality knob

$CT_0$ is the single parameter that trades quality for speed. It is the contrast at which the model declares detail invisible. Its effect on the reference display, at the recommended margin and floor:

| $CT_0$ | $f_c(0)$ | $B$ | **Primary-ray reduction** |
|---|---|---|---|
| 1/128 | 45.8 cpd | 0.0687 | 14.6× |
| **1/64** | **39.2 cpd** | **0.0539** | **18.6×** |
| 1/32 | 32.7 cpd | 0.0409 | 24.5× |
| 1/16 | 26.2 cpd | 0.0297 | 33.7× |

$CT_0 = 1/64$ is the conservative literature default and the recommended starting point. The staircase in §3.2 replaces it with a measured value.

## 4.4 From angles to the image plane

Two distinct angles are involved and conflating them is the most common error in this area.

- $e(x)$: **eccentricity**, the angle between the ray through $x$ and the gaze direction. Drives perception.
- $\theta(x)$: **off-axis angle**, the angle between the ray through $x$ and the camera axis. Drives geometry.

The perceptual requirement is angular: $2f_c(e)$ samples per degree. The renderer allocates samples per unit *image-plane area*. For a planar screen these differ by the projection Jacobian $d\Omega/dA = \cos^3\theta/D^2$. The primary-ray density, normalised so that an on-axis foveal location gets 1.0, is therefore

$$\boxed{D(x) = \max\left[\min\left(1,\ \frac{2f_c(e(x))}{N_d}\right)^2 \cos^3\theta(x),\quad D_{\min}\right]}$$

with the $\min(1,\cdot)$ enforcing the display Nyquist cap and $D_{\min}$ a floor (§4.6).

**Honest note on the Jacobian.** On this display it is worth including but it is not where the gain is. Aggregated over the frame it accounts for only 3.3% of the budget, because a 53° FOV keeps most of the screen area at modest $\theta$. Locally it reaches 1.53× at the corners, which is enough to cause a visible provisioning error there, and it costs one multiply. Include it, but do not expect it to move the headline number. (In a wide-FOV HMD the same term is worth a factor of three or more and becomes essential.)

## 4.5 Gaze safety margin

Following the expression in the physiology report, high-quality rendering extends beyond the nominal fixation point by

$r_{HQ} = r_{\text{task}} + k\sigma_g + v_{\text{eye}}\Delta.$

For an offline desktop renderer with a specified gaze point, $v_{\text{eye}}\Delta = 0$ and $\sigma_g$ reflects only the uncertainty in the specified fixation. For the gaze-tracked configuration (Stage C), a remote desktop tracker contributes accuracy ≈ 0.5° and precision $\sigma_g \approx 0.3°$, giving $m = r_{\text{task}} + 3\sigma_g \approx 1.5°$ during fixation.

Implementation: substitute $e' = \max(0, e - m)$ everywhere. Cost of the margin:

| $m$ | **0°** | **0.5°** | **1°** | **1.5°** | **2°** | **3°** | **5°** |
|---|---|---|---|---|---|---|---|
| Reduction | 26.3× | 23.5× | 20.9× | **18.6×** | 16.4× | 13.0× | 8.4× |

$m = 1.5°$ is the recommended operating point. Note how expensive latency is: at $m = 5°$ the system has already given away two thirds of its advantage, which is why the real-time version of this problem is hard and the offline version is not.

## 4.6 Floor and cap

The hyperbolic law sends density to zero at large eccentricity, which is wrong. Two guards:

- **Cap** at $D = 1$: never sample above display Nyquist. Already in the formula.
- **Floor** at $D_{\min} = 1/64$ (one primary ray per 8 × 8 display pixels). Justification: the CSF model addresses detection of static spatial detail and systematically over-predicts blur tolerance for large-field luminance, motion, and colour, all of which remain available in the far periphery. The floor binds beyond about 20° on this display and costs little (it raises $B$ from 0.0352 to 0.0380 at $m = 0$).

$D_{\min}$ is a free parameter that the §3.3 failure battery, not theory, should set.

---

# 5. Sample allocation

This section is the analytical core and it replaces the tuned exponents that heuristic treatments of this problem rely on.

## 5.1 The problem

Let the image be partitioned into sample sites $i$. Site $i$ covers screen area $A_i$, is assigned $n_i$ paths, and the per-path radiance variance there is $\sigma_i^2$. Let $w_i$ be the perceptual weight of squared error at site $i$. Minimise the perceptually weighted expected squared error subject to a total path budget:

$\min_{\{n_i\}} \sum_i w_i A_i \frac{\sigma_i^2}{n_i} \quad \text{s.t.} \quad \sum_i n_i = N.$

## 5.2 Solution

Lagrangian $\mathcal{L} = \sum_i w_i A_i \sigma_i^2 / n_i + \lambda(\sum_i n_i - N)$, so $-w_i A_i \sigma_i^2 / n_i^2 + \lambda = 0$, giving

$$\boxed{n_i \;\propto\; \sigma_i \sqrt{w_i A_i}}$$

This is Neyman allocation with a perceptual weight. It is not a new result in statistics; the point is that it is the *right* result here and it removes two free parameters. In particular the exponent on the variance is exactly 1, not a tunable $\alpha$, and the exponent on the perceptual weight is exactly $1/2$.

## 5.3 The consequence nobody expects

Substitute the terms. The perceptual weight is $w_i \propto \tau(e)^{-2}$ where $\tau(e) = \tau_0(1 + e/e_2)^{\gamma}$ is the tolerated error. The area per site is $A_i \propto 1/D(e)$, since thinning primary rays makes each surviving sample responsible for more screen area. With $D = \rho^2$ and $\rho = (1 + e/e_2)^{-1}$:

$$n(e) \;\propto\; \frac{\sigma}{\tau(e)\,\rho(e)} \;=\; \sigma\,\tau_0^{-1}\Big(1 + \tfrac{e}{e_2}\Big)^{1-\gamma}.$$

**If $\gamma < 1$, peripheral sites require more paths than foveal ones.**

The reason is physical, not perceptual. A noisy peripheral sample is smeared by reconstruction across the large area it represents, so its error contaminates more of the display than a noisy foveal sample does. The perceptual downweighting must overcome that area amplification before any per-sample saving appears at all.

This directly contradicts the standard intuition — "fovea 1024 spp, parafovea 256, periphery 64" — which is the configuration most first implementations reach for. That schedule corresponds to $\gamma \gg 1$ and is very likely to produce exactly the peripheral scintillation and blotching that foveated path tracers are known for. It is consistent with the observation in the literature that a linear acuity falloff still requires many rays in the periphery.

## 5.4 Why $\gamma$ is the critical parameter

Combining the two axes, the total path budget for the reference display at $m = 1.5°$, $CT_0 = 1/64$, floor 1/64, uniform $\sigma$, with per-site spp capped at 8× foveal:

| $\gamma$ | **Relative spp at $e = 20°$** | **Total path budget $T$** | **End-to-end reduction** |
|---:|---:|---:|---:|
| 0.00 | 8.0 (capped) | 0.193 | 5.2× |
| 0.25 | 4.6 | 0.130 | 7.7× |
| 0.50 | 2.7 | 0.090 | 11.2× |
| 0.75 | 1.5 | 0.064 | 15.7× |
| 1.00 | 0.9 | 0.047 | 21.3× |
| 1.50 | 0.3 | 0.028 | 35.4× |

A factor of seven between the extremes, on a parameter that heuristic designs assign by intuition. **Measuring $\gamma$ is the primary experimental objective of this project.** It is exactly what the §3.2 staircase measures if the staircase is run on $\gamma$ with $CT_0$ held fixed.

A prior expectation: $\gamma$ is small, probably in $[0, 0.5]$, because after foveated thinning the residual noise is already band-limited near the local cutoff $f_c(e)$, where the contrast threshold is by construction the same at every eccentricity. If that argument holds, the realistic end-to-end figure for this system is **5–11×**, not the 18.6× that the primary-ray budget alone suggests. The document should say so before measurement rather than after.

## 5.5 Variance estimation and the starvation trap

Driving allocation by *estimated* $\sigma_i$ has a known pathology: a site with few samples has a badly estimated variance, is therefore assigned few samples, and never recovers. Peripheral sites are exactly the low-sample sites. Mitigations, all required:

- **Warm-up.** Every site receives $n_{\min} \geq 16$ paths before any adaptive decision. Cycles' `adaptive_min_samples` already provides the hook.
- **Spatial smoothing of the variance field** before it enters the allocation, over a kernel scaled to the local site spacing (not a fixed pixel radius, which would be wrong by 8× between fovea and periphery).
- **Firefly handling.** Outlier radiance inflates $\sigma_i$ and, under Neyman allocation, attracts samples. That is correct behaviour for genuine caustics and wrong for a single spike. Clamp before estimating, trace after.

---

# 6. The sampling lattice

## 6.1 The constraint that decides this

A purely rendering-driven design would choose blue-noise-distributed sample positions realising $D(x)$, since blue noise gives the best spectral behaviour for a given density. That is the right answer for a monocular renderer and the wrong answer here.

The project's downstream goal is stereo matching on a non-uniform lattice. If the left and right images are thinned by two independent blue-noise realisations, corresponding scene points are almost never sampled in both images, and the row structure that every practical stereo matcher relies on is destroyed in both. That converts a hard research problem into an impossible one.

## 6.2 Row-structured lattices are free

The matcher's actual requirement is not separability of the density field, it is **row structure**: samples must lie on identifiable scanlines so that epipolar geometry in rectified coordinates remains a 1-D search.

This costs nothing. Any density field factors exactly as

$$D(u, v) = D_v(v) \cdot D_u(u \mid v), \qquad D_v(v) = \max_u D(u, v), \quad D_u(u \mid v) = \tfrac{D(u,v)}{D_v(v)} \leq 1,$$

where $D_v$ is the fraction of scanlines retained near row $v$ and $D_u$ the horizontal keep rate within a retained row. The realised total density is identical to the radial field by construction; verified numerically at $B = 0.0539$ for both.

For the reference display at the recommended operating point, this means **554 of 2160 scanlines retained (25.6%)**, with a mean horizontal keep rate of 21.0% on those rows:

| Row eccentricity | Rows retained | Mean horizontal keep |
|---:|---:|---:|
| 0° | 100% | 15.2% |
| 2° | 91.7% | 13.2% |
| 5° | 21.1% | 27.1% |
| 10° | 5.9% | 47.7% |
| 15° | 2.6% | 70.4% |

What *is* given up is isotropy: row-aligned samples produce anisotropic aliasing that blue noise would avoid. Mitigate by jittering horizontal positions within each row (free) and by dithering row selection with a 1-D blue-noise sequence over $v$ rather than a regular decimation (also free).

A warning on the tempting shortcut: making the density itself product-separable, $D(u, v) = \rho(e_u)\rho(e_v)$, does *not* save anything and costs 117% more budget than the radial field, because it produces cross-shaped rather than circular iso-density contours. Row structure is the requirement; separability of $D$ is not.

## 6.3 Stereo geometry

Fixation in stereo is a 3-D point, not a direction. For the reference display at 600 mm with 65 mm interocular separation the vergence angle is 6.2°, and each eye's gaze vector points at the fixation point $F$. Consequences:

- Eccentricity is computed **per eye**, from that eye's own gaze vector, not from a shared screen-space gaze point.
- Density is driven by the **cyclopean** eccentricity evaluated at the vergence depth, so that both eyes share the same $D_v(v)$ and therefore the same retained scanlines. Without this, the two images have different row sets and matching is dead on arrival.
- Within a row, the two eyes have different $D_u(u \mid v)$ warps, because a feature at disparity $d$ appears at a location of different local density in the other image. **This is the open research problem** (§12) and this document does not solve it; it only ensures the lattice design does not make it worse.

---

# 7. Reconstruction and denoising

## 7.1 Order of operations

The intuitive pipeline — upsample, then denoise — is wrong, and it is the most likely source of peripheral artefacts.

OpenImageDenoise and OptiX are trained on dense, approximately stationary-noise inputs. An image upsampled from spatially varying sample density violates both assumptions: the noise is nonstationary by construction, and the upsampling has already correlated neighbouring pixels. The denoiser will either over-smooth the fovea or leave low-frequency blotches in the periphery.

The correct order is **denoise in the sampling domain, then upsample**:

```
per-site radiance + albedo + normal + depth  (sparse, row-structured)
        │
        ▼
  denoise on the sample lattice        ← noise is stationary in sample count here
        │
        ▼
  edge-aware upsample to display grid  ← guided by full-resolution albedo/normal/depth
        │
        ▼
  full-resolution image
```

This is the same insight that motivates rendering and denoising in Visual-Polar space before mapping to screen space, arrived at from a different direction: their polar warp makes the sample grid regular so the denoiser sees stationary statistics. A row-structured lattice with per-row resampling achieves the same thing while preserving epipolar structure.

## 7.2 Auxiliary buffers

Albedo, normal, and depth must be available at **full display resolution** even where radiance is not, because they guide the upsampling. This is cheap: they are available from a primary-ray intersection with no shading and no secondary rays, so a full-resolution G-buffer pass costs a small fraction of a shaded sample. Budget it explicitly rather than pretending it is free; at roughly 2–4% of a full uniform render it is comparable to the entire foveated radiance budget and must appear in the accounting.

## 7.3 If a learned reconstructor is used

DeepFovea and successors reconstruct plausible peripheral content from very sparse input. That is a legitimate option but it changes the claim: the output is then a perceptual metamer, not a sample of the same light-transport solution. For a project whose stated discipline is holding light transport fixed, a learned reconstructor belongs in a clearly separated later experiment, and any image quality metric applied to it must be interpreted accordingly.

---

# 8. Gaze dynamics

Relevant only from Stage C onward, when a tracker is in the loop.

**Asymmetric transitions.** Promote to high resolution immediately when a region enters the foveal zone; demote over 150–300 ms when it leaves. Rationale: tracker noise at $\sigma_g \approx 0.3°$ would otherwise cause continuous level churn, and the cost of briefly over-rendering is small while the cost of exposing an under-rendered region after a saccade is a visible artefact.

**Saccadic suppression is the main lever and must not be neglected.** Saccades reach 500–900 °/s. At 20 ms end-to-end latency the term $v_{\text{eye}}\Delta$ demands 10–18° of high-quality margin, which from the table in §4.5 would destroy essentially the whole advantage. The standard escape is that visual sensitivity is strongly suppressed for roughly 50–100 ms around a saccade, so quality can be *reduced* mid-flight rather than pre-emptively raised. Detect saccade onset by velocity threshold, drop to the floor density during flight, and begin high-quality sampling at the predicted landing point before the eye arrives.

**What this implies for the offline system.** Nothing, directly — which is the point. Stages A and B specify gaze rather than tracking it, and therefore avoid the hardest part of the problem entirely. That is a deliberate scoping decision, not an oversight, and it should be stated as such.

---

# 9. Cycles integration

Targeting Cycles in Blender 5.x. Three notes correct common misconceptions about the current codebase.

**Tiles are not the scheduling unit any more.** Cycles X removed tiled GPU rendering in favour of full-frame progressive path tracing; tiles survive only for renders exceeding device memory, and measured optimal tile sizes on GPU are on the order of 1000², not 16² or 32². A design built around small tiles is targeting the pre-3.0 architecture. The modern analogue of tile-local sample correlation is **scrambling distance**, and it is worth investigating whether the foveation density field should modulate it — peripheral sites are sparse and spatially distant, so the coherence argument for low scrambling distance is weaker there.

**The real hook is the path-index mapping.** Cycles' GPU wavefront scheduler maps a linear path index to `(x, y, sample)` on the assumption that the render buffer is a dense array of pixels each accumulating equally-weighted samples. Variable primary-ray density breaks that assumption in two places: sites no longer correspond one-to-one with pixels, and accumulation must be area-weighted. The intervention points are the camera-ray generation kernel and the render-buffer accumulation, not a material node or a compositor trick.

**Per-pixel sample counts already exist.** Adaptive sampling maintains a per-pixel sample count and an auxiliary buffer for the convergence test, with `adaptive_min_samples` defaulting to the square root of the maximum sample count. That machinery is the natural carrier for the per-site $n_i$ of §5. What must change is that the noise threshold, currently a single global scalar in the range 0.1 to 0.001, becomes a lookup into the spatially varying $\tau(e)$ field.

## 9.1 Ordered list of code changes

1. **Foveation map upload.** A device buffer $F(u, v)$ carrying $\{D, \tau, n_{\min}\}$, computed on the host from gaze, camera, and display parameters. Read-only in the kernel.
2. **Primary-ray generation.** Replace the dense pixel iteration with a site list built from the row-structured lattice. Each site carries its represented area $A_i$ for weighted accumulation.
3. **Adaptive sampling threshold.** Replace the global scalar with a per-site lookup from $F$.
4. **Accumulation.** Area-weighted rather than uniform.
5. **Output.** Write a sparse radiance buffer plus full-resolution auxiliary G-buffer passes.
6. **Reconstruction.** Outside the kernel: denoise on the lattice, then guided upsample.

Steps 1, 3, and 5 are modest. Step 2 is the substantial one and is where the schedule risk sits.

---

# 10. Implementation stages

## Stage A — Reference implementation outside Cycles (4–6 weeks)

Pure Python plus Blender scripting, no Cycles modification. Render at several fixed resolutions over overlapping regions and composite with feathered masks. Deliberately inefficient — it does *more* work than a uniform render — but it produces images that are correct according to the model.

Purpose: run the §3.2 staircase and measure $CT_0$ and $\gamma$. **This is where the science happens.** Everything after it is engineering against measured parameters.

Deliverable: measured $CT_0$, $\gamma$, $D_{\min}$, with confidence intervals, and a FovVideoVDP-calibrated model.

### Stage B — Foveated Cycles, monocular (8–12 weeks)

The code changes of §9.1, single camera, gaze specified as a parameter. Validate that the measured $B$ and $T$ match the predicted 0.054 and the $\gamma$-dependent value from §5.4, and that JOD degradation stays under 1.0 at the parameters from Stage A.

Deliverable: a working foveated Cycles branch and a measured speedup curve.

### Stage C — Stereo and the non-uniform matcher (open-ended)

Extend to stereo pairs with a shared row structure (§6.3), then attack the matching problem. This is the research contribution and it should not be scheduled as if it were engineering.

---

## 11. Position relative to prior work

Nothing in §§4, 6, 7, 8 is new. It is a synthesis, and it should be presented as one.

- **Guenter et al. 2012**, *Foveated 3D Graphics* — three nested rasterised layers. Stage A is a path-traced version of this. Establishes the basic gain.
- **Patney et al. 2016**, *Towards Foveated Rendering for Gaze-Tracked VR* — showed that acuity-driven downsampling loses peripheral contrast and creates temporal aliasing, and that contrast enhancement raises the acceptable foveation level. Directly motivates §4.1.
- **Weier et al. 2016** — foveated ray tracing for HMDs with reprojection.
- **Koskela et al. 2016, 2019** — foveated path tracing, and Visual-Polar space, where rendering and denoising happen before the mapping to screen space, giving 2.5× on render and denoise at similar perceived quality with 1.3–1.5× faster traversal from improved primary-ray coherence. §7.1 is the same insight in a different coordinate system, and the coherence result is a real argument *against* the position taken here that should be tested, not dismissed.
- **Kaplanyan et al. 2019**, *DeepFovea* — learned reconstruction from sparse foveated samples. The alternative to §7 discussed in §7.3.
- **Tursun et al. 2019** — content-adaptive foveation, showing that a purely eccentricity-driven law ignores that displayed content strongly modulates sensitivity. A known limitation of this design.
- **Walton et al. 2021** — ventral metamers, the aggressive end of the peripheral-appearance spectrum.
- **Geisler & Perry 1998** — the model of §4.2.
- **Mitchell 1987; Overbeck et al. 2009; Rousselle et al.** — adaptive sampling and reconstruction, and the variance-estimation pathology of §5.5.
- **Mantiuk et al. 2021**, FovVideoVDP — the metric of §3.1.
- **Wang et al. 2022** (arXiv:2211.07969) — state-of-the-art survey; read before writing the related-work section.

### What is actually novel here

Three things, stated narrowly:

1. **Derived rather than tuned allocation.** The Neyman-with-perceptual-weight result of §5.2 and, more importantly, the consequence in §5.3 that peripheral sites may require more paths per sample. The literature allocates spp by eccentricity heuristics; this gives a criterion and predicts that the common heuristic has the sign wrong.
2. **Row-structured foveated lattices for stereo.** §6.2's observation that row structure is free — that any density field factors exactly into row-keep and within-row-keep rates — and §6.3's requirement of a shared cyclopean $D_v$ across eyes.
3. **Offline foveated path tracing as a distinct regime.** The literature is real-time and reports 2.5× against reconstruction-dominated baselines. Offline rendering amortises those costs and should get closer to the bound; nobody appears to have measured this.

Items 1 and 3 are modest but defensible. Item 2 only becomes a contribution if Stage C produces a working matcher.

---

# 12. Open problems

**The disparity-varying lattice phase.** Within a shared row, a feature at disparity $d$ falls at a location of different local sampling density in the two images. Correspondence therefore compares two differently-warped, differently-sampled 1-D signals. Possible directions: resample both rows into a common warped coordinate before matching; match in a scale-space whose scale is tied to local density; or make the density field depend on disparity itself, which is circular but may admit a fixed point. This is the project's real research question and this document does not answer it.

**Content dependence.** The model is content-blind. Tursun et al. show sensitivity depends strongly on what is displayed. A content-adaptive term could plausibly buy a further factor but interacts badly with the variance-driven allocation of §5, since both would then be estimating from the same noisy render.

**Whether $\gamma$ is stable across scenes.** §5.4 assumes a single $\gamma$. If it turns out to depend strongly on scene content, the allocation rule needs a per-scene calibration pass and the clean derivation loses much of its value.

**The Visual-Polar counter-argument.** Koskela's ray-coherence gain is real and this design gives it up in exchange for row structure. If the coherence penalty of a row-structured lattice turns out to be large, the stereo constraint may be better satisfied some other way. Measure it in Stage B rather than assuming.

---

# Appendix A — Parameter table

| Symbol | Meaning | Value / source |
|---|---|---|
| $D$ | Viewing distance | 600 mm (input) |
| $N_d$ | On-axis angular pixel density | 67.3 px/deg (derived) |
| $\alpha$ | CSF spatial-frequency decay | 0.106 (Geisler–Perry) |
| $e_2$ | CSF half-resolution eccentricity | 2.3° (Geisler–Perry) |
| $CT_0$ | Threshold contrast — quality knob | 1/64 initial; measure in Stage A |
| $m$ | Gaze safety margin | 1.5° |
| $D_{\min}$ | Density floor | 1/64; set by failure battery |
| $\gamma$ | Noise-tolerance exponent | **unknown; primary experimental target** |
| $\tau_0$ | Foveal noise threshold | Cycles default, 0.01 |
| $n_{\min}$ | Warm-up paths per site | ≥ 16 |
| $B$ | Primary-ray budget | 0.054 (derived) |
| $T$ | Total path budget | 0.028–0.193 depending on $\gamma$ |

# Appendix B — Reproducing the numbers

All figures in §§1, 4, 5, 6 come from a short script that builds the per-pixel eccentricity and off-axis fields for the reference display, evaluates $D(x)$, and integrates. It is worth keeping under version control alongside the report, since every number here changes if the display or viewing distance changes and none of them should ever be quoted without its configuration.